ECCO - Recovered Requirements

As of 2026-10-13.

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Purpose, scope and method

ECCO must let people manage a family of product variants as one repository: commit variants with the features they provide, learn which artifacts implement which features, and compose new variants for feature combinations never committed as such. The name stands for *Extraction and Composition for Clone-and-Own*.

These requirements were recovered, not written up front. ECCO has no requirements specification; it has about 1,400 commits since January 2016, a README, a changelog that stops at 0.1.9, and the research papers listed in the README. Each requirement below names where it was recovered from:

- **Doc** - stated in the README or a module README.
- **Code** - enforced by the implementation (a check, a refusal, a guard).
- **Test** - asserted by an automated test, the strongest evidence of intent.
- **History** - introduced or fixed by a specific commit; the fix shows what the system was expected to do.

Scope: the live Gradle modules (`base`, `service`, `logic`, `util`, `cli`, `gui`, `rest` and the adapters in `settings.gradle`). The retired storage backends, the `java6`/`java8`/`designspace` adapters and other unbuilt extras are out of scope.

shall = required and enforced; **should** = expected but not enforced everywhere.

Stakeholders and usage contexts

Stakeholder	Goal	Front end	Evidence
Variant developer (clone-and-own)	Keep developing individual variants while reusing features across them	CLI, GUI	README Use Cases; ICSME'14
Product-line engineer	Consolidate existing variants into a common platform; evolve features incrementally	GUI	SPLC'13, SoSyM'16, SST'15
Feature-location researcher	Compute feature-to-artifact traces from known variants; run benchmarks (SPLC challenge, VEVOS)	CLI, `experiment`, challenge adapter	SPLC'19
Distributed team	Exchange features, not whole histories, between repositories	CLI, GUI server	ICSE'16 doctoral symposium
Web-client user	Browse and operate repositories from a browser	REST + external ecco-client	`rest/README.md`

The use cases they bring:

1. **Feature location** - given variants with known configurations, compute which artifacts implement which feature.
2. **Extractive product-line engineering** - reverse-engineer a set of variants into a platform.
3. **Automated reuse for clone-and-own** - compose a new variant from features of existing ones.
4. **Reactive/incremental product-line engineering** - evolve variants and individual features over time.
5. **Feature-oriented distributed version control** - fork, pull and push with features as the unit of exchange.

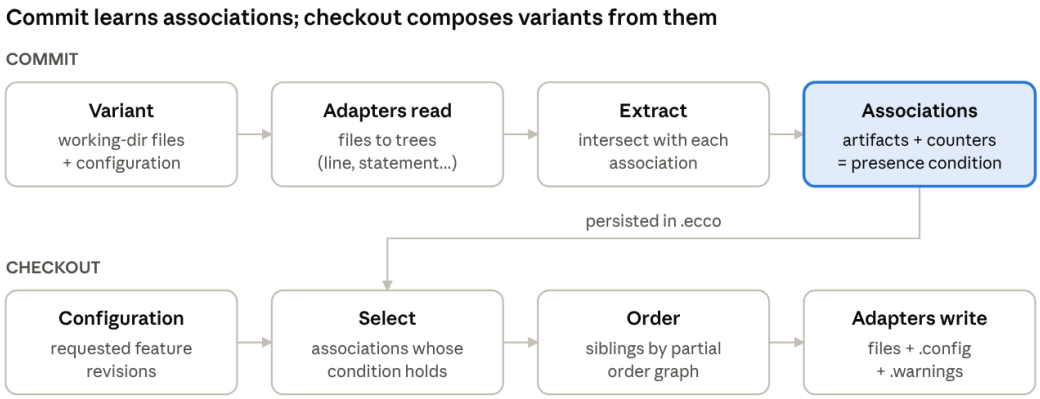
Use cases details them as step-by-step scenarios.

Domain model

The names are the interfaces in `base` (package `at.jku.isse.ecco`).

Term	Meaning	Code
Feature	A named unit of functionality; identified by id, the name need not be unique	`feature.Feature`
Feature revision	One version of a feature; `A'` creates a new one	`feature.FeatureRevision`
Configuration	The set of feature revisions a variant provides, e.g. `Base, Audio, Video.3f2a9c1`	`feature.Configuration`
Variant	A working directory's files together with its configuration; can be named and stored	`core.Variant`
Commit	One recorded variant: configuration, message, and the associations it touched	`core.Commit`
Artifact	One element of an artifact tree (file, line, statement, pixel…) with adapter-specific data	`artifact.Artifact`, `ArtifactData`
Node / artifact tree	The tree an adapter reads from files; nodes hold artifacts	`tree.Node`, `RootNode`
Association	A set of artifacts that always occurred together across commits	`core.Association`
Module / module revision	A feature interaction: positive and negative features (revisions) that co-occurred	`module.Module`, `ModuleRevision`
Counter	How often each module revision was seen with an association; the source of its condition	`counter.AssociationCounter`
Presence condition	Disjunction of module revisions; holds in a configuration when one of them holds	`module.Condition`
Partial order graph	The known orderings of sibling artifacts; ambiguous where commits disagree	`pog.PartialOrderGraph`
Feature trace	A presence condition stated up front (e.g. from VEVOS), not learned	`featuretrace.FeatureTrace`
Constraint	A mined or accepted relation between features (requires, excludes…)	`core.Constraint`
Remote	Another repository, by local path or `host:port`	`core.Remote`
Repository	All of the above, stored in a `.ecco` directory	`repository.Repository`

A commit never stores a variant whole: it splits the artifacts into associations, and a checkout reassembles any configuration whose associations' conditions hold.



Commit and checkout pipelines

Use cases

The use cases describe the key usage scenarios step by step, with their normal course and their alternatives. They follow RUCM (Restricted Use Case Modeling, Yue, Briand and Labiche, 2009). They are a complement to the requirements: the requirements say what must hold, the use cases say in which order actor and system interact. Each use case names the requirements it realizes.

Template. Each use case has a name, a brief description, a precondition, a primary actor, secondary actors, dependencies (INCLUDE USE CASE, EXTENDED BY USE CASE) and generalizations. After these come one basic flow and any number of alternative flows, and each flow ends with a postcondition. RUCM has three kinds of alternative flow:

- A **specific** flow (`RFS n`) branches at one step of the basic flow.
- A **bounded** flow (`RFS n–m`) can start at any of the steps n to m.
- A **global** flow can start at any step.

An alternative flow ends with `ABORT` (the use case ends) or `RESUME STEP n` (the flow continues at step n). This document adds two rows to the template: the front ends that offer the use case and the requirements it realizes.

Restrictions applied.

- Each step is one sentence in the simple present and the active voice, with the actor or the system as its subject.
- Steps use no modal verbs and no pronouns.
- There are four kinds of step: the actor requests, the system validates, the system acts internally, and the system replies.
- Conditions and loops use the RUCM keywords: `VALIDATES THAT`, `IF ... THEN ... ELSE ... ENDIF`, `DO ... UNTIL`, `MEANWHILE`.
- The system is written as “the system” whatever the front end.
- Messages in quotes are the system’s actual texts.

Actors

Actor	Kind	Stakeholder (see above)
Variant developer	Primary	Variant developer (clone-and-own)
Product-line engineer	Primary	Product-line engineer
Researcher	Primary	Feature-location researcher
Collaborator	Primary	Distributed team
Web-client user	Primary	Web-client user, through the external ecco-client
Remote repository	Secondary	Another ECCO repository, at a local path or behind a sync server (`host:port`)
LLM service	Secondary	Any OpenAI-compatible endpoint configured under Preferences
Git clone	Secondary	A local Git repository whose history is imported

Overview

ID	Use case	Primary actor	Front ends	Stakeholder use case
UC-1	Initialize repository	Variant developer	CLI, GUI	all
UC-2	Open repository	Variant developer	CLI, GUI	all
UC-3	Commit variant	Variant developer	CLI, GUI	1-4
UC-4	Commit several variants	Variant developer	GUI	1, 2
UC-5	Check out variant	Variant developer	CLI, GUI	3, 4
UC-6	Resolve ambiguous order	Variant developer	GUI	3, 4
UC-7	Locate features	Researcher	CLI, GUI	1
UC-8	Review mined constraints	Product-line engineer	GUI (CLI read-only)	2, 4
UC-9	Minimize presence conditions	Product-line engineer	CLI, GUI	2
UC-10	Import Git history	Product-line engineer	GUI	2, 4
UC-11	Fork repository	Collaborator	CLI, GUI, REST	5
UC-12	Exchange features with a remote	Collaborator	CLI, GUI, REST (pull only)	5
UC-13	Serve repository for synchronization	Collaborator	GUI	5
UC-14	Commit through the web client	Web-client user	REST	3
UC-15	Download variant through the web client	Web-client user	REST	3

The stakeholder use cases are the five listed under Stakeholders and usage contexts: 1 feature location, 2 extractive product-line engineering, 3 automated reuse for clone-and-own, 4 reactive product-line engineering, 5 feature-oriented distributed version control.

UC-1 Initialize repository

Brief description	The variant developer creates an empty repository for a directory.
Precondition	The session has no open repository.
Primary actor	Variant developer
Secondary actors	None
Dependency	None
Generalization	None
Front ends	CLI `init` (current directory); GUI Repositories > New
Realizes	FR-L1, FR-L2, FR-L4

Basic flow

1. The variant developer requests a new repository for a directory.
2. The system VALIDATES THAT the directory holds no repository.
3. The system creates the `.ecco` directory.
4. The system stores the defaults from `ecco.properties` in the repository.
5. The system opens the new repository.

Postcondition: An empty repository exists in `<directory>/.ecco` and is open.

Specific alternative flow SA1 (RFS 2)

1. The system reports “Repository already exists at this location.”.
2. ABORT.

Postcondition: The existing repository is unchanged.

Bounded alternative flow BA1 (RFS 3-5)

1. IF the creation of the repository fails THEN the system removes the half-created repository ENDIF.
2. The system reports the error with its cause.
3. ABORT.

Postcondition: The directory holds no repository, and the variant developer can retry.

UC-2 Open repository

Brief description	The variant developer opens an existing repository, which every other use case except UC-1 and UC-11 needs.
Precondition	None
Primary actor	Variant developer
Secondary actors	None
Dependency	None
Generalization	None
Front ends	CLI (each command opens the repository itself); GUI Repositories > Open
Realizes	FR-L3, FR-L4, FR-L5, FR-P2, FR-P4, IF-C8, NFR-V1

Basic flow

1. The variant developer requests to open a directory.
2. The system VALIDATES THAT the directory exists and holds repository data.
3. The system VALIDATES THAT exactly one storage plugin and at least one enabled artifact adapter are available.
4. The system VALIDATES THAT the repository format is supported.
5. The system loads the repository.
6. The system enables the repository operations.

Postcondition: The repository is open, and the system has written nothing.

Specific alternative flow SA1 (RFS 1)

1. IF the variant developer runs a CLI command THEN the system searches the current directory and its parents for the nearest `.ecco` ENDIF.
2. The system uses the directory that holds `.ecco` as the working directory.
3. RESUME STEP 2.

Postcondition: The repository directory is known.

Specific alternative flow SA2 (RFS 2)

1. The system reports that the path holds no repository, with a hint at a nested `.ecco` where one exists.
2. ABORT.

Postcondition: The system has written nothing.

Specific alternative flow SA3 (RFS 3)

1. The system reports the missing storage plugin or the missing enabled adapter.
2. ABORT.

Postcondition: The session has no open repository.

Specific alternative flow SA4 (RFS 4)

1. The system reports the older or retired format with an explanation.
2. ABORT.

Postcondition: The repository is unchanged.

UC-3 Commit variant

Brief description	The variant developer records the files of a working directory as a variant with a configuration; the system splits the artifacts into associations and updates their presence conditions.
Precondition	A repository is open (UC-2). The working directory holds the variant's files.
Primary actor	Variant developer
Secondary actors	None
Dependency	Included by UC-4, UC-7, UC-10, UC-14
Generalization	None
Front ends	CLI `commit –c <configuration> [–m <message>]` (prints the commit id and any violated constraints); GUI Versions > Commit
Realizes	FR-F1 to FR-F6, FR-M1 to FR-M11, FR-P1 to FR-P3, FR-T1, AD-11, IF-C3, IF-C9, NFR-R1, NFR-R3

Basic flow

1. The variant developer enters a configuration and a commit message for the working directory.
2. IF the configuration is empty THEN the system reads the configuration from `.config` in the working directory ENDIF.
3. The system VALIDATES THAT the configuration is well-formed and names at least one feature.
4. The system begins a read-write transaction.
5. The system reads every file that `.ecco/.ignores` does not exclude, with the adapter that `.ecco/.adapters` maps the file to.
6. The system VALIDATES THAT the repository holds no artifacts in a retired adapter format.
7. The system intersects the artifacts with every association.
8. The system updates the counters of every association that the commit touches.
9. IF the configuration is new THEN the system records the configuration as a variant ENDIF.
10. The system commits the transaction.
11. The system VALIDATES THAT the configuration satisfies the accepted constraints.
12. The system reports success.

Postcondition: The repository holds the commit, and the presence condition of every touched association reflects the configuration. A later checkout of the same configuration reproduces the files (NFR-R3).

Specific alternative flow SA1 (RFS 3)

1. The system reports the configuration error: invalid syntax, an ambiguous feature name, an unknown feature id, or "A commit needs at least one feature: … (e.g. BASE).".
2. ABORT.

Postcondition: The repository is unchanged.

Specific alternative flow SA2 (RFS 6)

1. The system rolls back the transaction.
2. The system reports that the repository holds a retired adapter format and can only be checked out.
3. ABORT.

Postcondition: The repository is unchanged.

Specific alternative flow SA3 (RFS 11)

1. The system reports each violated constraint as a warning.
2. RESUME STEP 12.

Postcondition: The commit is stored; the violation is advisory (FR-K8).

Specific alternative flow SA4 (RFS 5)

1. IF the adapter of a file cannot read the file THEN the system reads the file with the text adapter if the file is text, else with the file adapter ENDIF.
2. The system logs a warning that names the file and the error.
3. RESUME STEP 6.

Postcondition: The file is committed with the fallback adapter and checks out byte for byte (AD-11).

Bounded alternative flow BA1 (RFS 4-10)

1. IF a file is unreadable, a file maps to a disabled or missing adapter, an adapter cannot run (for example its Python modules are missing), storing fails, or the commit runs out of memory THEN the system rolls back the transaction ENDIF.
2. The system reports the error with its cause; running out of memory is reported with the heap size and how to raise it.
3. ABORT.

Postcondition: The repository is in its state before the commit (NFR-R1).

UC-4 Commit several variants

Brief description	The variant developer commits several variant folders of one parent folder in a chosen order, typically to consolidate existing clones into a repository.
Precondition	A repository is open (UC-2). Each variant folder holds one variant, usually with a `.config`.
Primary actor	Variant developer
Secondary actors	None
Dependency	INCLUDE USE CASE Commit variant
Generalization	None
Front ends	GUI Versions > Commit Multiple Versions
Realizes	IF-G2, FR-M3

Basic flow

1. The variant developer selects a parent folder.
2. The system VALIDATES THAT the parent folder has subfolders.
3. The variant developer selects the variant folders.
4. The system lists the selected folders with the configuration from each folder's `.config`, on one line.
5. The variant developer orders the folders and edits configurations and commit messages.
6. The system shows the constraint violations of each configuration.
7. The variant developer requests the commit.
8. The system VALIDATES THAT at least one folder is listed.
9. The system VALIDATES THAT no configuration violates an accepted constraint.
10. DO
11. INCLUDE USE CASE Commit variant for the next folder with the folder's configuration and message.
12. The system logs the folder and the commit time.
13. UNTIL every listed folder is committed.
14. The system shows the log and the last commit.

Postcondition: The repository holds one commit per listed folder, in the listed order.

Specific alternative flow SA1 (RFS 2)

1. The system reports "The selected folder has no subfolders".
2. RESUME STEP 1.

Postcondition: The folder list is unchanged.

Specific alternative flow SA2 (RFS 8)

1. The system reports "Add at least one folder to commit.".
2. RESUME STEP 1.

Postcondition: Nothing is committed.

Specific alternative flow SA3 (RFS 9)

1. The system lists the violations and asks for confirmation.
2. IF the variant developer confirms THEN RESUME STEP 10 ELSE RESUME STEP 5 ENDIF.

Postcondition: Nothing is committed yet.

Bounded alternative flow BA1 (RFS 11-12)

1. IF a commit fails THEN the system shows the error next to the log ENDIF.
2. ABORT.

Postcondition: The folders committed before the failure stay committed; the remaining folders are not committed.

UC-5 Check out variant

Brief description	The variant developer composes the variant for a configuration, which may be a feature combination never committed as such, and writes it into an empty working directory.
Precondition	A repository is open (UC-2) and holds at least one commit.
Primary actor	Variant developer
Secondary actors	None
Dependency	EXTENDED BY USE CASE Resolve ambiguous order; included by UC-15
Generalization	None
Front ends	CLI `checkout -c <configuration> [--minimized]`; GUI Versions > Checkout, View > Variants, View > Artifacts (compose a selection)
Realizes	FR-K1 to FR-K10, FR-K12, FR-N9, FR-T2, AD-6, IF-C9

Basic flow

1. The variant developer enters a configuration for a working directory.
2. The system VALIDATES THAT the configuration is well-formed.
3. The system VALIDATES THAT the working directory is empty apart from `.ecco`.
4. The system begins a read-only transaction.
5. The system selects the associations whose presence condition holds for the configuration.
6. The system prunes the composed artifact tree by the repository's evaluation strategy.
7. The system orders sibling artifacts by the partial order graphs.
8. The system collects MISSING, SURPLUS, ORDER, UNRESOLVED, CONSTRAINT and TRACE warnings.
9. The system ends the transaction.
10. The system writes the files with the adapters.
11. The system writes `.config` and `.warnings` into the working directory.
12. The system shows the warnings with a suggested fix for each. EXTENDED BY USE CASE Resolve ambiguous order.

Postcondition: The working directory holds the composed variant, `.config` and `.warnings`; the repository is unchanged.

Specific alternative flow SA1 (RFS 2)

1. The system reports the configuration error.
2. ABORT.

Postcondition: The working directory is unchanged.

Specific alternative flow SA2 (RFS 3)

1. The system reports that the working directory is not empty or already holds `.config`, `.warnings` or `.hashes`.
2. ABORT.

Postcondition: The working directory is unchanged.

Specific alternative flow SA3 (RFS 5)

1. IF the variant developer requested minimized conditions THEN the system selects with the stored minimized conditions that are still valid and with the full conditions for the other associations ENDIF.
2. RESUME STEP 6.

Postcondition: The selection equals the selection with full conditions for every configuration that the trusted constraints allow (FR-N9).

Specific alternative flow SA4 (RFS 12)

1. IF the variant developer works with the CLI THEN the system prints the configuration and the number of warnings of each kind, with a pointer to `.warnings` ENDIF.
2. ABORT.

Postcondition: Same as the basic flow.

UC-6 Resolve ambiguous order

Brief description	After a checkout reports an ORDER warning, the variant developer chooses the order of an artifact's children, and the system records the order in the repository without a commit, so that later checkouts know the order.
Precondition	A checkout in the GUI (UC-5) reports an ORDER warning.
Primary actor	Variant developer
Secondary actors	None
Dependency	Extends UC-5 at step 12
Generalization	None
Front ends	GUI checkout details; CLI `order <file>...`
Realizes	IF-G8, IF-C10, FR-K2, FR-K5, FR-K11

Basic flow

1. The variant developer selects the ORDER warning and requests a reorder.
2. The system shows the children of the artifact, with the pairs whose order earlier commits fixed shown as locked.
3. The system VALIDATES THAT at least one child can move.
4. The variant developer moves children into the intended order.
5. The variant developer confirms the order.
6. The system VALIDATES THAT the order agrees with every order recorded before.
7. The system records the order in the partial order graph of the artifact.
8. The system rewrites the file in the checkout directory in the new order.
9. The system shows the warnings without the resolved ORDER warning.

Postcondition: The repository records the chosen order, and later checkouts that contain these children use it, whatever their configuration. The repository holds no new commit or variant, and the presence conditions and the other warnings are unchanged.

Specific alternative flow SA1 (RFS 3)

1. The system reports "Nothing to reorder: these children's relative order is already fully determined by prior commits.".
2. The variant developer closes the dialog.
3. ABORT.

Postcondition: The checkout directory and the repository are unchanged.

Bounded alternative flow BA1 (RFS 4-5)

1. IF the variant developer cancels the dialog THEN ABORT ENDIF.

Postcondition: The checkout directory and the repository are unchanged.

Specific alternative flow SA2 (RFS 6)

1. The system reports that the order contradicts one recorded before.
2. ABORT.

Postcondition: The checkout directory and the repository are unchanged.

Specific alternative flow SA3 (RFS 1)

1. IF the variant developer works with the CLI THEN the variant developer puts the content of the checked-out file in the intended order ENDIF.
2. The variant developer runs `order` with the file.
3. The system composes the configuration of `.config` again.
4. The system reads the file with the adapter.
5. IF the file holds content the checkout does not, or lacks content of it, THEN the system reports that content and ABORT ENDIF.
6. The system prints the number of precedences recorded.
7. RESUME STEP 6.

Postcondition: As the basic flow, and the file is already in the new order; after an ABORT the repository is unchanged.

UC-7 Locate features

Brief description	The researcher commits variants with known configurations and inspects which artifacts implement which feature or feature interaction.
Precondition	A repository is open (UC-2). The researcher has a set of variants with known configurations, for example a VEVOS or SPLC challenge benchmark.
Primary actor	Researcher
Secondary actors	None
Dependency	INCLUDE USE CASE Commit variant

Generalization	None
Front ends	CLI `traces [id]`, `dg`; GUI View > Associations, View > Artifacts, association preview, Visualize > Dependency Graph
Realizes	FR-M1, FR-T1, FR-T2, IF-C6, IF-G6, IF-G7, IF-G9

Basic flow

1. DO
2. INCLUDE USE CASE Commit variant for the next variant.
3. UNTIL every variant is committed.
4. The researcher requests the associations.
5. The system lists each association with its full and simplified presence condition.
6. The researcher selects associations.
7. The system shows the artifacts of the selected associations in their files, coloured by association.

Postcondition: The researcher has the feature-to-artifact traces; the repository holds the committed variants.

Specific alternative flow SA1 (RFS 2)

1. IF a variant holds VEVOS presence conditions (`pcs.variant.csv`) and the C, C++ or challenge adapter reads the variant THEN the system records the presence conditions as proactive feature traces ENDIF.
2. RESUME STEP 3.

Postcondition: The proactive traces are stored with their associations; a trace that contradicts the commits is not used, and checkout reports the trace as a TRACE warning (FR-T2).

Specific alternative flow SA2 (RFS 4)

1. IF the researcher passes an association id to `traces` THEN the system prints the association's presence condition and artifact tree ENDIF.
2. ABORT.

Postcondition: The repository is unchanged.

Specific alternative flow SA3 (RFS 4)

1. The researcher requests the dependency graph.
2. The system writes the association dependency graph as GML (CLI) or shows the graph for export (GUI).
3. ABORT.

Postcondition: The repository is unchanged.

UC-8 Review mined constraints

Brief description	The product-line engineer reviews feature constraints mined from the committed configurations and accepts or rejects them, which builds up a feature model.
Precondition	A repository is open (UC-2) and holds commits with different configurations.
Primary actor	Product-line engineer
Secondary actors	None
Dependency	INCLUDE USE CASE Minimize presence conditions
Generalization	None
Front ends	GUI View > Feature Model; CLI `suggest-constraints`, `accept-constraint`, `reject-constraint`
Realizes	FR-N1 to FR-N6, IF-C5, IF-G5

Basic flow

1. The product-line engineer opens the feature model.
2. The system mines MANDATORY, REQUIRES and EXCLUDES constraints per feature from the committed configurations, with the minimum witness count (default 4) and the confidence (default 0.9).
3. The system shows the pending suggestions, which are the mined ones minus the accepted and the rejected ones, hard ones first.
4. The product-line engineer selects suggestions and accepts the selection.
5. The system stores the accepted constraints in the repository in one transaction.
6. INCLUDE USE CASE Minimize presence conditions.
7. The system shows each accepted constraint with the constraint's trust state.

Postcondition: The repository holds the accepted constraints, which travel with fork, pull and push. An accepted constraint is trusted only while re-mining still yields the constraint.

Specific alternative flow SA1 (RFS 3)

1. The product-line engineer changes the minimum witness count or the confidence.

2. RESUME STEP 2.

Postcondition: The suggestions match the new thresholds.

Specific alternative flow SA2 (RFS 4)

1. The product-line engineer rejects the selected suggestions.
2. The system stores the rejections in the repository in one transaction, withdrawing an acceptance of the same suggestion.
3. RESUME STEP 3.

Postcondition: The rejected suggestions are no longer proposed, here or in any repository that receives them by fork, pull or push.

Specific alternative flow SA3 (RFS 4)

1. The product-line engineer moves accepted constraints back to pending.
2. The system removes the constraints from the repository in one transaction.
3. INCLUDE USE CASE Minimize presence conditions.
4. RESUME STEP 3.

Postcondition: The constraints are suggestions again.

Specific alternative flow SA4 (RFS 1)

1. IF the product-line engineer runs `suggest-constraints` THEN the system prints the pending suggestions and how many are accepted and rejected; with `--all` it also lists the accepted ones with their trust state and the rejected ones ENDIF.
2. IF the product-line engineer runs `accept-constraint` or `reject-constraint` with a kind and its features THEN the system stores the decision as in step 5 or SA2, and with `--undo` moves the suggestion back to pending as in SA3; a feature the repository does not have, or the wrong number of features for the kind, is refused ENDIF.
3. ABORT.

Postcondition: The repository holds the decision; `suggest-constraints` alone leaves it unchanged.

Specific alternative flow SA5 (RFS 1)

1. IF this machine still holds rejections of an earlier version for the repository THEN the system moves the rejections into the repository, except for suggestions the repository has accepted ENDIF.
2. RESUME STEP 2.

Postcondition: The rejections are stored in the repository and no longer on this machine.

UC-9 Minimize presence conditions

Brief description	The product-line engineer simplifies the presence conditions of all associations under the accepted feature model, for reading and optionally for checkout.
Precondition	A repository is open (UC-2).
Primary actor	Product-line engineer
Secondary actors	None
Dependency	Included by UC-8
Generalization	None
Front ends	CLI `minimize`, `minimize-preview`; GUI “Minimize Presence Conditions”
Realizes	FR-N7, FR-N8, FR-N9, IF-C5

Basic flow

1. The product-line engineer requests the minimization.
2. The system compiles the trusted accepted hard constraints into a feature model.
3. The system minimizes the presence condition of every association, dropping a literal or term only where a SAT check proves the result equivalent under the feature model.
4. The system stores the minimized conditions together with the inputs they were computed from.
5. The system reports the number of associations with stored minimized conditions.

Postcondition: The minimized conditions are stored and are valid until the association’s condition, the committed configurations or the accepted constraints change. A checkout with minimized conditions (UC-5 SA3) uses them.

Specific alternative flow SA1 (RFS 1)

1. IF the product-line engineer requests a preview (`minimize-preview`) THEN the system shows the original and the minimized condition of each association ENDIF.
2. ABORT.

Postcondition: The repository is unchanged.

Specific alternative flow SA2 (RFS 1)

1. IF a minimization is already running THEN the system ignores the request ENDIF.
2. ABORT.

Postcondition: The running minimization continues.

Bounded alternative flow BA1 (RFS 2-4)

1. IF the minimization fails THEN the system reports the error ENDIF.
2. ABORT.

Postcondition: The previously stored minimized conditions stay as they were.

UC-10 Import Git history

Brief description	The product-line engineer turns the history of a Git repository into ECCO commits, one per Git commit, choosing a configuration for each with help from an LLM.
Precondition	A repository is open (UC-2). A local Git clone exists.
Primary actor	Product-line engineer
Secondary actors	Git clone, LLM service
Dependency	INCLUDE USE CASE Commit variant
Generalization	None
Front ends	GUI Versions > Import From Git
Realizes	FR-G1 to FR-G3, FR-G4, IF-G3, IF-G4

Basic flow

1. The product-line engineer selects a local Git clone.
2. The system VALIDATES THAT the folder holds a `.git`.
3. The system shows the history along first parents, oldest first.
4. The product-line engineer selects a range of commits, the review interval N and the use of LLM suggestions.
5. DO
6. The system extracts the tree of the next Git commit into an empty directory.
7. The system VALIDATES THAT the commit is due for review.
8. The system sends the diff, the message and the features imported so far to the LLM service.
9. The system VALIDATES THAT the LLM service returns a suggestion.
10. The system shows the commit with a configuration made of the running feature set and the suggestion.
11. The product-line engineer edits the configuration.
12. The product-line engineer chooses Import.
13. The system VALIDATES THAT the configuration satisfies the accepted constraints.
14. INCLUDE USE CASE Commit variant for the extracted tree.
15. The system makes the committed configuration the running feature set.
16. UNTIL the last selected commit is processed.
17. The system shows the log, the last commit and a summary of the warnings.

Postcondition: The repository holds one commit per imported Git commit. The working tree and the index of the Git clone are unchanged.

Specific alternative flow SA1 (RFS 2)

1. The system reports “Not a git repository (no .git found)”.
2. RESUME STEP 1.

Postcondition: Nothing is imported.

Specific alternative flow SA2 (RFS 7)

1. The system sends the commit to the LLM service.
2. IF the LLM service returns no suggestion THEN the system logs the failure ENDIF.
3. The system takes the running feature set and the suggestion as the configuration.
4. RESUME STEP 14.

Postcondition: The commit is imported without review; a commit that fails is shown for review (SA7).

Specific alternative flow SA3 (RFS 9)

1. The system shows the commit with the running feature set and the reason the suggestion failed.
2. RESUME STEP 11.

Postcondition: The product-line engineer completes the configuration by hand.

Specific alternative flow SA4 (RFS 12)

1. The product-line engineer chooses Skip.
2. RESUME STEP 16.

Postcondition: The Git commit is not imported, and the running feature set is unchanged.

Specific alternative flow SA5 (RFS 12)

1. The product-line engineer chooses Stop.
2. The system logs “Stopped: imported X of Y commit(s).”.
3. ABORT.

Postcondition: The commits imported so far stay in the repository.

Specific alternative flow SA6 (RFS 13)

1. The system asks whether to import the commit despite the constraint violation.
2. IF the product-line engineer confirms THEN RESUME STEP 14 ELSE RESUME STEP 11 ENDIF.

Postcondition: Nothing is committed yet.

Specific alternative flow SA7 (RFS 14)

1. IF the commit fails THEN the system logs the failure ENDIF.
2. The system shows the commit with the configuration that was tried and the reason for the failure.
3. RESUME STEP 11.

Postcondition: Nothing of the failed commit is in the repository; the product-line engineer corrects the configuration and imports again, or skips the commit.

UC-11 Fork repository

Brief description	The collaborator creates a new repository from a remote one, optionally leaving out feature revisions.
Precondition	The session has no open repository.
Primary actor	Collaborator
Secondary actors	Remote repository
Dependency	None
Generalization	None
Front ends	CLI `fork <remote> [--exclude ...]`; GUI Repositories > Fork; REST fork with deselected features
Realizes	FR-D1, FR-D2, FR-D4, FR-D5, FR-D7, FR-L2, FR-N4, IF-R2

Basic flow

1. The collaborator enters a remote and the feature revisions to exclude.
2. The system VALIDATES THAT the remote is a local path or a valid `host:port`.
3. The system VALIDATES THAT the target location holds no repository.
4. The system reads the remote repository’s features, associations and constraints.
5. The system VALIDATES THAT the exclusion leaves no unresolved dependencies.
6. The system drops the features left without revisions and the modules that contain excluded features.
7. The system stores the result as the new repository.
8. The system registers the remote as `origin`.

Postcondition: The new repository holds the remote’s content without the excluded revisions; reopened, the new repository checks out the same content as the remote. The remote repository is unchanged (FR-D7).

Specific alternative flow SA1 (RFS 2)

1. The system reports “Invalid remote address provided.”.
2. ABORT.

Postcondition: Nothing is created.

Specific alternative flow SA2 (RFS 3)

1. The system reports “A repository already exists at the given location”.
2. ABORT.

Postcondition: The existing repository is unchanged.

Specific alternative flow SA3 (RFS 5)

1. The system removes the half-created repository.
2. The system reports “Unresolved dependencies in selection.”.
3. ABORT.

Postcondition: Nothing is created.

Bounded alternative flow BA1 (RFS 4-8)

1. IF the connection or the copy fails THEN the system removes the half-created repository ENDIF.
2. The system reports the error, for example “Error connecting to remote: …”.
3. ABORT.

Postcondition: Nothing is created, and the collaborator can retry.

UC-12 Exchange features with a remote

Brief description	The collaborator pulls features from a remote into this repository or pushes features from this repository into the remote, optionally leaving out feature revisions; fetch only reads the remote’s feature list.
Precondition	A repository is open (UC-2). The remote is registered, by default as `origin`.
Primary actor	Collaborator
Secondary actors	Remote repository
Dependency	None
Generalization	None
Front ends	CLI `fetch`, `pull`, `push [remote] [--exclude ...]`; GUI Collaborate > Remotes; REST pull between its own repositories
Realizes	FR-D1, FR-D3, FR-D4, FR-D5, FR-D7, FR-M9, FR-N4, IF-C4, IF-G10, IF-R4

Basic flow

1. The collaborator requests a pull from a remote, with the feature revisions to exclude.
2. The system VALIDATES THAT the remote is registered.
3. The system reads the remote repository.
4. The system VALIDATES THAT the exclusion leaves no unresolved dependencies.
5. The system drops the excluded revisions and the modules that contain the excluded features.
6. The system VALIDATES THAT this repository holds no artifacts in a retired adapter format.
7. The system merges the selection into this repository in one transaction.
8. The system reports success.

Postcondition: This repository holds the union of both repositories’ selected content, with accepted constraints merged without duplicates; reopened, this repository checks out the pulled content as the remote does. The remote repository is unchanged.

Specific alternative flow SA1 (RFS 1)

1. The collaborator requests a push to the remote instead.
2. The system reads this repository.
3. The system VALIDATES THAT the exclusion leaves no unresolved dependencies.
4. The system merges the selection into the remote repository.
5. RESUME STEP 8.

Postcondition: The remote repository holds the union of both repositories’ selected content; this repository is unchanged.

Specific alternative flow SA2 (RFS 1)

1. The collaborator requests a fetch instead.
2. The system reads the remote’s features.
3. The system stores the features with the remote’s entry.
4. ABORT.

Postcondition: The associations of both repositories are unchanged.

Specific alternative flow SA3 (RFS 2)

1. The system reports “Remote ‘’ does not exist.”.
2. ABORT.

Postcondition: Both repositories are unchanged.

Specific alternative flow SA4 (RFS 4)

1. The system rolls back the transaction.
2. The system reports “Unresolved dependencies in selection.”.
3. ABORT.

Postcondition: Both repositories are unchanged.

Specific alternative flow SA5 (RFS 6)

1. The system reports that the repository holds a retired adapter format and can only be checked out.
2. ABORT.

Postcondition: Both repositories are unchanged.

Bounded alternative flow BA1 (RFS 3-7)

1. IF the connection to the remote fails THEN the system rolls back the transaction ENDIF.
2. The system reports “Error connecting to remote: …”.
3. ABORT.

Postcondition: Both repositories are unchanged.

UC-13 Serve repository for synchronization

Brief description	The collaborator makes the open repository available to other ECCO instances for fork, fetch, pull and push.
Precondition	A repository is open (UC-2).
Primary actor	Collaborator
Secondary actors	Remote repository (the connecting ECCO instance)
Dependency	None
Generalization	None
Front ends	GUI Preferences, server section
Realizes	FR-D6, IF-G10, NFR-S1, NFR-S3

Basic flow

1. The collaborator enters a port (default 3770).
2. The collaborator requests the start of the server.
3. The system VALIDATES THAT no server is running.
4. The system listens on the port of the loopback interface.
5. The system shows “Server running on port P” with a log.
6. DO
7. A remote repository sends a fork, fetch, pull or push request.
8. The system answers the request, deserializing only allow-listed classes.
9. The system logs the time of the connection and the request.
10. UNTIL the collaborator stops the server.
11. The system stops the server.

Postcondition: The server is stopped. The repository holds what remote repositories pushed.

Specific alternative flow SA1 (RFS 3)

1. The system reports “Server is already running.”.
2. ABORT.

Postcondition: The running server is unchanged.

Specific alternative flow SA2 (RFS 4)

1. IF the collaborator selected “Accept connections from other machines” THEN the system warns that synchronization has no authentication ENDIF.
2. IF the collaborator confirms THEN the system listens on the port of all interfaces ELSE ABORT ENDIF.
3. RESUME STEP 5.

Postcondition: Any machine that reaches the port can read and push into the repository.

UC-14 Commit through the web client

Brief description	The web-client user uploads the files of a variant with a configuration, and the server commits them.
Precondition	The REST server runs with a JWT secret and a users file. The web-client user has an account.
Primary actor	Web-client user

Secondary actors	None
Dependency	INCLUDE USE CASE Commit variant
Generalization	None
Front ends	REST `POST /login`, `POST /api/{repository}/commit/add`
Realizes	IF-R1, IF-R3, IF-R6, NFR-S2, NFR-S4, NFR-S5

Basic flow

1. The web-client user sends a user name and a password.
2. The system VALIDATES THAT the credentials match a user.
3. The system returns a signed bearer token.
4. The web-client user uploads files, a commit message and a configuration for a repository.
5. The system VALIDATES THAT the repository exists.
6. The system VALIDATES THAT the configuration is well-formed and names at least one feature.
7. The system VALIDATES THAT every file name stays inside the repository's upload folder.
8. The system replaces the contents of the upload folder with the files.
9. INCLUDE USE CASE Commit variant for the upload folder, with the web-client user as committer.
10. The system returns the updated repository.

Postcondition: The repository holds the commit.

Specific alternative flow SA1 (RFS 2)

1. The system reports "Invalid user name or password".
2. ABORT.

Postcondition: The web-client user holds no token.

Specific alternative flow SA2 (RFS 5)

1. The system returns 404, "repository with the id does not exist".
2. ABORT.

Postcondition: Nothing is written.

Specific alternative flow SA3 (RFS 6)

1. The system returns 400 with the configuration error.
2. ABORT.

Postcondition: Nothing is written.

Specific alternative flow SA4 (RFS 7)

1. The system returns 400, "Invalid path: …".
2. ABORT.

Postcondition: Nothing is written.

Global alternative flow GA1

1. IF a request after step 3 carries no valid signed token THEN the system returns 401 ENDIF.
2. ABORT.

Postcondition: The repository is unchanged.

UC-15 Download variant through the web client

Brief description	The web-client user downloads a stored variant as a zip file.
Precondition	The web-client user holds a token (UC-14 steps 1-3). The repository has a named variant.
Primary actor	Web-client user
Secondary actors	None
Dependency	INCLUDE USE CASE Check out variant
Generalization	None
Front ends	REST `GET /api/{repository}/variant/{variant}/checkout`
Realizes	IF-R5, IF-R6

Basic flow

1. The web-client user requests the checkout of a variant.
2. The system VALIDATES THAT the repository exists.
3. The system creates a fresh folder.
4. INCLUDE USE CASE Check out variant for the variant's configuration into the folder.
5. The system zips the folder.
6. The system deletes the folder.
7. The system returns the zip file.

Postcondition: The web-client user has the variant's files with `.config` and `.warnings`; the repository is unchanged.

Specific alternative flow SA1 (RFS 2)

1. The system returns 404.
2. ABORT.

Postcondition: Nothing is written.

Bounded alternative flow BA1 (RFS 3-5)

1. IF the folder or the zip file cannot be created THEN the system returns 500 ENDIF.
2. ABORT.

Postcondition: The repository is unchanged.

Global alternative flow GA1

1. IF the request carries no valid signed token THEN the system returns 401 ENDIF.
2. ABORT.

Postcondition: The repository is unchanged.

Functional requirements

Paths are abbreviated: `ES` = `service/.../service/EccoService.java`, `REPO` = `base/.../repository/Repository.java`, `DR/DW` = `service/.../adapter/dispatch/DispatchReader.java/DispatchWriter.java`, `STS` = `service/.../storage/ser/dao/SerTransactionStrategy.java`, `SR` = `service/.../storage/ser/repository/SerRepository.java`.

Repository lifecycle

ID	Requirement	Source
FR-L1	`init` shall create `.ecco` and refuse if the service is already initialized or the directory exists; defaults come from `ecco.properties` (max order 2, PROACTIVE evaluation, BOOST main tree, `ser` storage).	Code ES:870-895
FR-L2	A failed `init` or `fork` shall remove the half-created repository so the operation can be retried.	Test `FailedForkCleanupTest`
FR-L3	`open` shall refuse a missing path, a non-directory or a directory without repository data, write nothing, and hint at a nested `.ecco`.	Test `OpenNonRepositoryTest`, `ForkFromWorkingDirectoryTest`; History f8cad7e1
FR-L4	The service shall require exactly one storage plugin and at least one enabled artifact adapter.	Code ES:386-424
FR-L5	Every operation shall fail with “Service is not initialized” while no repository is open.	Code ES:375

Configurations

ID	Requirement	Source
FR-F1	A configuration shall be a comma-separated list of `A` (latest revision, or new feature), `A'` (new revision), `A.<rev>` (that revision) or `[<id>]` (feature by id); names are limited to `[a-zA-Z0-9_-]`.	Doc; Code `Configuration.java:27`, `ConfigurationParser`
FR-F2	A revision id prefix of 7 or more characters shall resolve if unique.	Test `ShortRevisionIdTest`; History f2e97cc2
FR-F3	An ambiguous feature name shall be rejected with a hint to use the feature id.	Code `ConfigurationParser:175`
FR-F4	Parsing a configuration shall not persist revisions as a side effect.	Test `ParseConfigurationPhantomRevisionTest`
FR-F5	A configuration shall hold at most one revision per feature.	Code REPO:509-516
FR-F6	`[<id>]` shall name an existing feature, with the same meaning as its name in every form; an unknown id is rejected.	Test `UnknownFeatureIdTest`

Commit

ID	Requirement	Source
FR-M1	A commit shall read the working directory through the adapters, extract its artifacts into associations, and update each association’s counters (and so its presence condition).	Doc; Code `CommitService`, `Repository.extract`
FR-M2	A commit without any feature shall be refused.	Code `CommitService:55`; History 4e150250
FR-M3	Without an explicit configuration, a commit shall use `<working dir>/.config`, and fail if it exists but cannot be read. Line breaks in the file shall be collapsed into one line (`a.1, b.1`), which is also the default commit message.	Code `CommitService.getConfigStringFromFile`; Test `ConfigFileLineBreaksTest`
FR-M4	Each previously unseen configuration shall be recorded as a variant.	Code `CommitService:70-95`
FR-M5	Files matching `.ecco/.ignores` (defaults `.DS_Store`, `.gitignore`) shall be skipped; `.ecco`, `.config`, `.warnings` and `.hashes` are always skipped.	Code DR:31-39; Test `DsStoreIgnoredRegressionTest`, `GitignoreIgnoredRegressionTest`
FR-M6	Files shall be routed by `.ecco/.adapters` (`pluginId;glob`, first match wins), written on first open from the adapters’ priorities and fixed for the repository from then on.	Doc; Code DR:143-197
FR-M7	A file mapped to a disabled or missing adapter shall fail the commit, not fall back to another adapter; malformed `.adapters` lines are rejected. (A file its enabled adapter fails to parse does fall back, AD-11.)	Test `UnavailableAdapterRoutingTest`
FR-M8	An unreadable file shall fail the commit, not be committed empty; symlink loops are skipped.	Test `UnreadableFileCommitTest`, `SymlinkLoopCommitTest`
FR-M9	A commit or merge shall be refused into a repository holding artifacts in a retired adapter format; checkout still works.	Code REPO:1049-1090, `ArtifactData#retiredFormat`
FR-M10	A commit shall run in one read-write transaction and roll back on any failure, also on an `Error` such as `OutOfMemoryError`; a listener failure never rolls back a committed transaction. Every other operation that opens a transaction, read-only or read-write, shall roll it back on any failure too, and pass an `Error` on unwrapped.	Code `CommitService`, `ListenerRegistry:76-82`; Test `ErrorRollbackTest`; History 2dc75b70

ID	Requirement	Source
FR-M11	A file whose adapter, path, content (SHA-256) and context are the same as in the session's previous read should not be parsed again: its tree from then is copied, and the commit stores the same as with a fresh read. Adapters opt in by naming their context (`ArtifactReader#readContext`): C, C++, TypeScript, Python, Java AST, Markdown, LilyPond; text and plain files are read as before. – `Decco.reuseReadTrees=false` turns it off. `.hashes` is not used for this - see Known gaps #3.	Code DR, `Trees#copyRead`; Test `DispatchReaderReuseTest`; History 90135e96

Checkout and composition

ID	Requirement	Source
FR-K1	A checkout shall select the associations whose presence condition holds for the configuration and prune the composed tree per node by the repository's evaluation strategy (RETROACTIVE, PROACTIVE, PROACTIVE-ADDITION, PROACTIVE-SUBTRACTION).	Code REPO:756-763, `CheckoutComposer`, `featuretrace/evaluation`
FR-K2	Sibling order shall come from the partial order graph; a graph with cycles, redundant transitive edges or wrong node counts is rejected.	Code `DefaultOrderSelector`, `PartialOrderGraph:347–358`
FR-K3	A checkout shall write into a working directory that is empty apart from `.ecco`, and refuse if `.config`, `.warnings` or `.hashes` exist.	Code DW:80-106, `CheckoutService:120–136`
FR-K4	A checkout shall write `.config` (the configuration) and `.warnings`.	Doc; Code `CheckoutService:139–167`
FR-K5	`.warnings` shall list, one per line with a suggested fix where one exists: MISSING feature interactions never committed, SURPLUS artifacts the configuration does not explain, ambiguous ORDER, UNRESOLVED associations, CONSTRAINT violations, and rejected TRACEs.	Code `CheckoutService:139–167`; Test `CheckoutMissingModuleDiagnosticsTest`
FR-K6	MISSING warnings shall be trimmed: an interaction is not reported when a smaller sub-combination of it is already missing.	Code `ModuleRevisions.java:38–58`
FR-K7	SURPLUS warnings shall be absorbed (X + XY = X) and suppressed when entailed by the accepted feature model; both on by default, a failure only logs, MISSING warnings are never suppressed.	Code `SurplusLatticeAbsorber`, `SurplusModuleSuppressor`; History ce32c243
FR-K8	Constraint-violation warnings shall be advisory and never block a commit or checkout.	Code ES:1196-1207
FR-K9	Composition shall run in a read-only transaction.	Test `ReadWithoutTransactionTest`
FR-K10	Line-based files shall be written with the newest commit's encoding and line separator; whether a file ends with a newline shall follow the variant checked out (AD-6).	Test `TextFormatLastCommitWinsTest`, `FinalNewlineVariantsTest`; History 8282a32d
FR-K11	The order of an artifact's children shall be recordable without a commit: only the artifact's partial order graph changes, the order must agree with every order recorded before and is refused otherwise, and it holds in every later checkout containing those children, whatever the configuration.	Test `RecordOrderTest`, `OrderResolutionCharacterizationTest`, `OrderCommandTest`
FR-K12	A file removed in a later commit (for example by renaming its directory) shall not be checked out in variants that do not contain it: a node that only leads to its association's own nodes contributes no condition of its own to the main tree.	Code `SerNode.copySingleNodeCompletely`, `Trees.treeFusion`; Test `RemovedFileCheckoutTest`; History d73f17e0

Constraints and condition minimization

ID	Requirement	Source
FR-N1	The system shall mine MANDATORY, REQUIRES and EXCLUDES constraints from committed configurations, with a minimum witness count (>= 1, default 4) and a confidence threshold (0-1, default 0.9; 1.0 = hard rules only), sorted hard first, then by witnesses.	Code `ConstraintMiner:80–210`
FR-N2	Mined constraints shall be suggestions only: never applied unless accepted.	Code `ConstraintMiner`
FR-N3	Constraints shall be mined per feature, so revisions of one feature are never mutually exclusive.	Code `ConfigurationBridge`
FR-N4	Accepted and rejected constraints shall be persisted in the repository and travel with fork, pull and push without duplication; a suggestion is accepted, rejected or neither, and on a merge the receiving repository keeps its own decision. Rejections kept per machine by earlier versions move into the repository.	Test `ConstraintPersistenceMergeTest`, `RejectedConstraintsTest`
FR-N5	An accepted constraint shall only be trusted while re-mining still yields it.	Test `AcceptedConstraintStaleReMineTest`
FR-N6	Accepting, rejecting or returning many constraints shall be one transaction and one event.	Code `ConstraintService:84–122`; History 04825b26
FR-N7	Presence-condition minimization shall drop a literal or term only when SAT-proven equivalent under the accepted feature model.	Code `PresenceConditionMinimizer`; Doc (`minimize-preview` is read-only)
FR-N8	A stored minimized condition shall only be shown or used while the association's condition, the distinct committed configurations and the accepted constraints it was computed from are unchanged.	Test `MinimizedConditionValidityTest`
FR-N9	On request (`checkout --minimized`, GUI preference; off by default), checkout shall use the stored revision-exact minimized conditions that are still valid, and write the same files as without them for every configuration the trusted accepted constraints allow.	Test `MinimizedCheckoutEquivalenceTest`, `MinimizeCommandTest`

Proactive feature traces

ID	Requirement	Source
FR-T1	The C, C++ and challenge adapters shall read VEVOS presence conditions (`pcs.variant.csv`) as proactive feature traces, using feature names that match the configuration's.	Code `VevosConditionHandler`; History f258f279
FR-T2	A proactive trace shall hold in every commit containing its association and in no other; a contradicting trace is not used and is reported as a TRACE warning (too narrow / too broad) with the commit.	Code `ProactiveTraceCheck`; History 980daf85

Distributed operations

ID	Requirement	Source
FR-D1	A remote shall be a local path (repository or its `.ecco`) or `host:port` (hostname, IPv4 or bracketed IPv6, port 1-65535); remote names are unique.	Test `RemoteAddressTest`
FR-D2	`fork` shall create a new repository from a remote into a location that holds no repository, and register it as `origin`.	Code ES:663-800
FR-D3	`fetch` shall read a remote's features and store them with the remote's entry; `pull` and `push` shall merge a remote into this repository or this one into the remote; all three report an unknown remote as "Remote '' does not exist.".	Doc; Code `RemoteSyncService`; Test `RemoteSyncCharacterizationTest`
FR-D4	`fork`, `pull` and `push` shall be able to exclude feature revisions: features left without revisions and modules containing excluded features are dropped, and an exclusion that leaves unresolved dependencies is refused.	Doc; Code REPO:867-1033
FR-D5	A fork or pull, reopened, shall check out the same content as the source.	History 2c299089, 09fa089a
FR-D7	A fork, fetch or pull shall not modify the remote repository, also when it excludes revisions or fails.	Test `ForkLeavesOriginUnchangedTest`, `FetchPullLeaveRemoteUnchangedTest`
FR-D6	The sync server shall refuse a second start, and accept only allow-listed classes when deserializing.	Test `RemoteSyncServerHardeningTest`; Code `SyncObjectStreams`
FR-D8	`fork`, `pull` and `push` shall merge only files both repositories read the same way (`.ecco/.settings`): a fork takes its origin's settings, also over the network; a pull or push of a plugin's files into a repository with neither that plugin's settings nor its files takes the sender's over; otherwise differing settings refuse the operation before anything is merged, naming them. A server too old to tell its settings is synced with as before.	Test `LilypondSettingsSyncTest`

Git import

ID	Requirement	Source
FR-G1	Git import shall read a local clone without touching its working tree or index, and apply commits oldest first along first parents.	Code `git/GitHistoryReader`
FR-G2	Each Git commit's tree shall be extracted to an empty directory, skipping submodules and refusing entries outside it.	Code `GitHistoryReader`
FR-G3	LLM suggestions shall be one request per commit (temperature 0, JSON output) judged against features already imported; a failure gives a blank suggestion, never an abort, and a suggestion is never committed without the import loop's decision.	Code `LlmFeatureSuggestionClient:398–410`
FR-G4	A Git commit that fails to import shall be shown again for review with the error and the configuration tried, to be corrected and imported again or skipped; the run goes on and nothing of the failed commit is kept.	Test `ImportGitViewRetryTest`

Persistence (`ser` storage)

ID	Requirement	Source
FR-P1	A repository shall be stored in `.ecco` as a core file (`<id>.ser.zip`, the current one named by the file `id`: features, constraints, modules, commits, the ids of the associations, the list of artifact packs in the order they were written and each pack's size), one file per association in `associations/`, and the artifacts in packs in `artifacts/`.	Code STS, SR
FR-P2	The core shall hold no entry per artifact. Loading shall read the packs in the order they were written, a later pack's copy of an artifact superseding an earlier one's, and rebuild in memory which pack holds each artifact's current copy.	Code SR (`packOrder`), STS; uncommitted 2026-10-13
FR-P3	A commit shall write one new pack with only the artifacts whose serialized bytes differ from their stored ones (compared by a digest kept per artifact id); a pack that keeps fewer than half of its artifacts current is repacked into the new one, and above 20 packs the smallest are merged.	Code STS `writeArtifactPack`; Test `ArtifactPackTest`, `UnchangedArtifactRewriteTest`
FR-P4	A core written before packs were ordered (it lists each artifact and its pack, and no pack order) shall be read the old way; its next write shall put every current artifact into one new pack and leave the per-artifact index out of the core, once.	Code SR `hasLegacyArtifactIndex`, STS; Test `OldStorageFormatTest` (loading only); measured 2026-10-13

Artifact adapter requirements

An adapter reads files of one kind into artifact trees and writes them back. The finer its trees, the more precisely features are traced. Fidelity differs by adapter: the line-based adapters (text, Markdown, C, C++) are byte-exact, and so are TypeScript and LilyPond for a variant checked out as committed; the Java (AST) and Python adapters are not.

Contract for every adapter

ID	Requirement	Source
AD-1	An adapter shall be a `ArtifactPlugin` registered through `ServiceLoader`, binding its reader, writer and optional viewers through Guice; `getPluginId()` returns the plugin class name.	Code; `adapter/README.md`; per-adapter plugin-id tests
AD-2	A reader shall return one root node per file carrying `PluginArtifactData`; a writer is chosen by that plugin id.	Code DR, DW
AD-3	A reader shall declare prioritized glob patterns; higher priority wins when `.adapters` is first written, and a tie between enabled adapters is broken by plugin id and logged.	Code; Test `DispatchReaderJavaPluginPriorityTest`, `DispatchReaderPriorityTieTest`
AD-4	Adapters shall be enabled or disabled per user; at least one must be enabled.	Code `AdapterPreferences`, ES
AD-5	Viewers shall not be bound when running headless (`ecco.headless`).	Code; History bfcddc23
AD-6	Line-based adapters shall round-trip byte-exact: encoding (UTF-8, else ISO-8859-1), line separator and final newline. The final newline shall be traced per variant (`FinalNewlineArtifactData`, the last child of the file's node); encoding and line separator are per file, the newest commit's. This holds for the text, Markdown, C and C++ adapters; a file stored before keeps the final newline recorded for it then.	Tests `TextFileFormatTest` (5,000 random rounds), `FinalNewlineVariantsTest`; History 8282a32d
AD-7	An adapter whose tree format changes shall mark old data with `retiredFormat()` or a tree-format check, so old repositories are refused for commit, not corrupted.	Code `ArtifactData#retiredFormat`, `JavaASTData.TREE_FORMAT`, `cpp/data/RetiredFormat`
AD-8	An adapter that needs Python shall fail the commit when Python or its modules are missing, not commit an empty file, and shall pick a free py4j port.	Test `PythonGatewayPortTest`, `LilypondGatewayPortTest`
AD-9	An adapter that needs Python shall find its interpreter, not rely on `python` on the PATH: the one set in Preferences if any, else the first of `python`, `python3` and the usual install locations (MacPorts, Homebrew, python.org, `/usr/local/bin`, `/usr/bin`) that imports its modules - libcst and py4j for Python; tree-sitter, py4j and lymodel for LilyPond (lymodel's tree-sitter parse reads the files), lymodel from the lymodel directory in Preferences, `LYPYTHON` or installed. When none will do, the error shall name each interpreter tried and what it lacks; a module installed after a failure shall be found without restarting.	Code `PythonFinder`, `LilypondPython`, `PY4JParser.python()`; Test `PythonFinderTest`, `LilypondPythonTest`; History d8c8ac60, 6765398d, 143e32cb
AD-10	The LilyPond adapter shall read with lymodel's tree-sitter parse, recorded in a new repository's settings (`lilypond.parser=lybar-2`). A repository without that setting was read with parce, whose tokens and contexts the parse would never match, and one with `lybar-1` holds tokens that do not record whether they were written together with the one before them (part of a token's identity since `lybar-2`): their files shall still check out, but a commit shall be refused with an explanation before anything is written. A repository with no LilyPond settings and no LilyPond files takes the settings of a new repository at its next commit.	Code `LilypondReader#readable`, `EccoService#adoptRepositorySettings`, `LilypondPlugin.PARSER_SETTING`, `LilypondParser_2.py`; Test `LilypondRepositorySettingsTest`, `LilypondParseTest`; History 4b95ee97, 143e32cb, lybar-2 2026-10-11
AD-11	A file its adapter cannot read shall not fail the commit: it shall be read by the text adapter if it is text (no NUL byte in its first 8 KB), else by the file adapter, both byte-exact, with a warning naming the file and the error (a `.json` with comments, a Git LFS pointer named `.png`, a construct an adapter does not know). An adapter that cannot run at all (no Python, a repository made for another version of it: `AdapterUnavailableException`) and a cancelled commit shall still fail it.	Test `DispatchReaderFallbackTest`; History 86ea679c

Per adapter

Adapter	Files	Granularity	Fidelity required	On by default	Evidence
File	everything else	whole file	byte-exact	yes	`FileReaderTest`, `FileWriterTest`
Text	`.txt` `.xml` `.html` `.css` `.js` `.java`	lines	byte-exact	yes	`TextReaderTest`, `TextFileWriterTest`
Markdown	`.md` `.markdown`	CommonMark/GFM blocks, nested by heading	byte-exact	yes	`MarkdownReaderTest`, `MarkdownFileWriterTest`, `MarkdownRoundTripPropertyTest` (2,000 generated documents)

Adapter	Files	Granularity	Fidelity required	On by default	Evidence
Image	`.png` `.jpg` `.jpeg` `.bmp` `.gif`	pixels (one artifact each: a 4-megapixel PNG takes 230 s, 9 GB and 1 GB stored per commit; README "Images" maps images to the file adapter)	exact colours for PNG/BMP; JPEG lossy; never an empty file	yes	`ImageFormatWriteTest`
Java (AST)	`.java`	JavaParser AST, Java 21 level	every comment kept, also inside lambdas and expressions; braces, `synchronized`, arrow cases kept; layout pretty-printed, not byte-exact	yes	`JavaASTCommentTest`, `JavaASTStatementFidelityTest`, `JavaASTTreeFormatTest`
C	`.c` `.h`	lines grouped by function; `#if` as lines; VEVOS traces	byte-exact incl. blank lines, CRLF, Latin-1	yes	`CRoundTripTest`
C++	`.cpp` `.hpp` (+ `.c` `.h` when C is off)	lines grouped by namespace, class, enum, function	byte-exact incl. include guards and comments; first format checkout-only	yes	`CppRoundTripTest`, `CppRoundTripPropertyTest` (2,000 generated files), `LegacyCppWriterTest`
TypeScript	`.ts`	statements and blocks (TS compiler in embedded Node.js)	exact text incl. `;`, trailing comments, JSDoc, declarations without a body or initializer (`.d.ts`, overloads), empty blocks, enums and classes; switch/enum order kept - 161 real files (28 of flipperzero-firmware) byte for byte	yes	`TypeScriptRoundTripTest`, `TypeScriptOrderTest`
Python	`.py` `.ipynb` `.json`	libcst syntax tree; notebook cells; JSON values	not byte-exact: whitespace-only lines normalized, JSON re-indented, notebook outputs ignored	yes	`PythonAdapterTest`
LilyPond	`.ly` `.ily`	tokens from lymodel's tree-sitter parse (parce before `lilypond.parser=lybar-1`, AD-10; since `lybar-2` a token records whether it was attached to the one before it); with musical tokens each note one token carrying its meaning, and with variable contexts each top-level assignment a context of its name	byte-exact incl. tabs, CRLF and blank lines when checked out as committed: a token keeps the whitespace it was read with after the token before it (not part of its identity) - all 724 corpus and example files with musical tokens (the default), and all 705 corpus files read without them; where a checkout puts a token after another one, the writer's rules space it, and the music is the same (judged by lymusic: all 719 files it reads); tokens stored before keep the writer's rules	yes	`LilypondParseTest`, `LilypondVariantsCommitCheckoutTest`
Go	`.go`	tokens (ANTLR)	exact reconstruction	**no**	`GoWriterTest`
Runtime	`.runtime` `.java`	Java classes/methods/lines + btrace data	not format-preserving	**no**	`RuntimeWriterTest`
Java (lines)	`.java`	class, imports, members, statements	read-only: checkout refused	**no**	`JavaWriterTest`
Challenge	`.java`	class, method, line + VEVOS traces	read-only: checkout refused	**no**	`ChallengeReader*Test`

Interface requirements

Three front ends sit on one service API (`EccoService`): a CLI for scripting and experiments, a JavaFX GUI for exploration, and a REST server for the external web client. They do not offer the same operations; the matrix at the end shows where they differ.

Command line (`cli`)

ID	Requirement	Source
IF-C1	The CLI shall offer `init`, `status`, `adapters`, `commit -c [-m]`, `checkout -c`, `features`, `traces`, `get/set`, `remotes`, `fork`, `fetch`, `pull`, `push`, `dg`, `suggest-constraints`, `accept-constraint`, `reject-constraint`, `minimize-preview`, `minimize` and `order`; `checkout` takes `--minimized`.	Code `Main.registerCommands()`
IF-C2	`commit` and `checkout` shall require a configuration (`-c`).	Code
IF-C3	Exit codes shall be 0 on success, 1 on a failed command, 2 on invalid arguments; errors go to stderr with their cause chain, the stack trace only with `-Decco.debug=true`. The error shown shall be the command's own: a failure to close the repository afterwards is attached to it, not shown in its place; running out of memory is reported as such, with the heap size and how to raise it (`JAVA_OPTS=-Xmx16g`).	Code `Main.run`, `Command#withOpenRepository`; Test `CommitCommandTest`; History 2dc75b70
IF-C4	`fetch`, `pull` and `push` shall default to the remote `origin`; `pull`, `push` and `fork` take `--exclude` feature revisions.	Code
IF-C5	`suggest-constraints` and `minimize-preview` shall take `--min-witness` (default 4) and `--confidence` (default 0.9); only hard (confidence 1.0) accepted constraints are ever applied. `suggest-constraints` shall leave out accepted and rejected suggestions unless given `--all`.	Code
IF-C6	`dg` shall print the association dependency graph as GML on stdout.	Code
IF-C7	The CLI shall run without JavaFX and bundle the same adapters as the GUI, so it can work on repositories made with the GUI.	Code (`ecco.headless`, openjfx excluded); Doc
IF-C8	The CLI shall find the repository in the current or nearest parent directory and use that directory as the working directory; `init` and `fork` create a repository in the current directory.	Doc; Test `MainRepositoryDiscoveryTest`
IF-C9	`commit` shall print the commit id and configuration, then each violated accepted constraint as `CONSTRAINT:`; `checkout` shall print the configuration and the number of warnings of each kind, pointing to `.warnings`.	Test `CommitCommandTest`, `CheckoutCommandTest`
IF-C10	`order <file>...` shall record the order of checked-out files as they are now, without a commit (FR-K11): the configuration comes from `.config`, the files are matched to its composition by content, content added or removed is refused, and paths are relative to the directory the command runs in.	Test `OrderCommandTest`, `RecordOrderTest`

Graphical user interface (`gui`)

ID	Requirement	Source
IF-G1	The GUI shall group its functions in a ribbon: Repositories (new, open, fork, close), Versions, Collaborate, View, Visualize; repository actions are disabled until a repository is open, and new, open and fork while one is.	Code `MainView`, `RibbonBar`
IF-G2	Users shall be able to commit one variant folder or several at once, check out configurations, and manage named variants (add, remove, edit feature revisions, check out several into a base directory).	Code `CommitBaseDirView`, `CommitView`, `VariantsView`
IF-G3	Users shall be able to import a Git history commit by commit, oldest first, with Import, Skip and Stop per reviewed commit, auto-importing the commits between every N-th review (N 1-1000, default 1).	Code `ImportGitView`
IF-G4	Git import should suggest a configuration per commit from an LLM at any OpenAI-compatible endpoint, configured under Preferences (URL, model).	Code `PreferencesView`; Doc
IF-G5	The feature model view shall show the feature model derived from the commits and let users accept, reject or return mined constraints, and minimize presence conditions with them.	Code `FeaturesView`, `ConstraintSuggestionsView`
IF-G6	Users shall be able to list and compare commits, list associations with their full and simplified conditions, and browse the artifact tree of any selection of associations, then check out or compose that selection.	Code `CommitsView`, `CommitComparisonView`, `AssociationsView`, `ArtifactsView`
IF-G7	Wherever associations are listed, an association preview shall show their artifacts in their files, coloured by association, for every mapped adapter.	Doc; Code `ArtifactDetailView` viewers
IF-G8	Checkout results shall show their warnings with a suggested fix; an ambiguous order shall be resolvable by reordering in a dialog, which records the order without a commit (FR-K11).	Code `CheckoutDetailView`, `ReorderChildrenDialog`; Test `CheckoutDetailViewRecordOrderTest`
IF-G9	The GUI shall visualize a knowledge graph (features, commits, variants, associations), the artifact graph and the dependency graph, plus charts, each exportable.	Code `view/graph/*`, `ChartsView`
IF-G10	The GUI shall manage remotes, fetch, pull and push, and start a sync server on a chosen port, local-only unless "accept connections from other machines" is set.	Code `RemotesView`, `ServerView`

ID	Requirement	Source
IF-G11	Users shall be able to enable or disable adapters (effective on the next open), and set minimization thresholds, LilyPond paths, and the Python the Python and LilyPond adapters run (and LilyPond's lymodel directory) when it is not found (AD-9).	Code `PreferencesView`

REST server (`rest`)

ID	Requirement	Source
IF-R1	The REST server shall serve the repositories of a storage directory (`ECCO_STORAGE_DIR`) to the external ecco-client on port 8081 (`PORT`), with an OpenAPI description and Swagger UI.	Doc; Code `application.yml`; Test `SettingsTest`
IF-R2	It shall let clients list, create, clone, fork (deselecting features) and delete repositories.	Code `RepositoryController`
IF-R3	It shall accept a commit as a multipart upload of files, message, configuration and user name, and refuse a malformed configuration or one without features with 400 before writing any file.	Code `CommitController`; Test `FileRepositoryServiceTest`
IF-R4	It shall let clients describe features and feature revisions, and pull features from another repository while deselecting some.	Code `FeatureController`
IF-R5	It shall let clients manage variants (add, rename, add/update/remove features) and download a variant's checkout as a file.	Code `VariantController`
IF-R6	Every endpoint except Swagger shall require an authenticated JWT bearer token.	Code `@Secured(IS_AUTHENTICATED)`

Where the front ends differ

Operation	CLI	GUI	REST
Commit / checkout	yes	yes	commit upload / variant download
Record an order (ORDER warnings)	`order <file>...`	Reorder…	no
Named variants	no	yes	yes
Fork	yes	yes	yes
Fetch / pull / push	yes	yes	pull only, between its own repositories
Sync server	no	yes	no
Constraint mining	suggest, accept, reject, preview, minimize	suggest, accept, reject, minimize	no
Git import	no	yes	no
Visualizations	`dg` as GML	yes	no

Non-functional requirements

None of these were written as targets; they are recovered from refusals in the code and from fixes that treated a behaviour as a defect. The baselines are measurements recorded with those fixes, not promises.

Reliability and data integrity

ID	Requirement	Source
NFR-R1	A commit shall be atomic: a crash leaves the previous state or the new one, never a mix. Dirty files are staged and rolled forward after the id swap.	Doc; History a6f4acac, feb839d1
NFR-R2	Durability shall not rely on per-file fsync; atomicity comes from the roll-forward journal.	History: fsync measured ~10x slower (112 s vs 10.6 s, 21k artifacts) and rejected
NFR-R3	A commit followed by close, reopen and checkout shall reproduce exactly what was committed, for repositories, forks and pulls alike.	Test `CommitCheckoutTest`; History 2c299089, 09fa089a
NFR-R4	Distinct artifacts with equal content shall stay distinct through persistence.	History c288ed2d, da14de14
NFR-R5	Errors shall surface, not be dropped: missing associations throw instead of being skipped; a rollback never follows a successful commit.	History c6bb7561
NFR-R6	Write-time pruning of associations shall not be used; it loses data on fork-and-merge.	History (reproduced, closed)

Performance (recorded baselines)

ID	Requirement	Baseline	Source
NFR-P1	Commit and checkout should scale to tens of features and variants interactively.	60 variants / 40 features: commit 1.8 s, checkout 1.1 s, status 0.2 s	History cd764f4e, 81442a9e
NFR-P2	Opening a repository should take seconds, not minutes.	Real x8 repo: 65 s to 5.3 s	History decb2e37
NFR-P3	A commit shall rewrite only what changed.	15.7 s to 2.6 s after skipping identical writes. Still rewritten: every association a commit touches (#48), and every touched artifact is serialized to find out whether it changed (#49)	History 4bb0961c
NFR-P4	Pixel-granular images should be usable at around 1.7 megapixels.	71k-px image: commit 67 to 2 s, checkout 157 to 3.4 s	History 7a4e0aec, 47f6f0e5
NFR-P5	Storage should stay compact as commits accumulate.	Association files 147 to 13 MB (compact counters); packs 2.5-7x smaller	History 81442a9e, 7c18da73
NFR-P6	Merging a file into its partial order graph should not be quadratic in its length; every commit merges every file, changed or not.	Large Git import: one commit 73 s to 19 s, another out of memory to 35 s; a 47-commit LilyPond repository replays to the same checkouts and stored content	History 3c9ce88f (`AlignPathsEquivalenceTest`)
NFR-P7	A file unchanged since the session's previous read should not be parsed again (FR-M11).	15 flipperzero-firmware commits: 238 s to 205 s, a typical commit 13 s to 10 s	History 90135e96
NFR-P8	What a commit writes should not grow with the number of artifacts in the repository (FR-P2).	flipperzero-firmware: core written per commit 26.5 MB to ~0; a commit after the first 10.9 s to 8.9 s by comparing artifacts by id, to 8.3 s with the index left out of the core; the first commit 23.5 s to 19.5 s	Uncommitted 2026-10-13

Partial-order-graph alignment is factorial in the number of concurrent unresolved branches; above 100,000 orderings a fast fallback takes over, so commits stay fast (`ConcurrentBranchScalingTest`). Aligning two long paths builds an n x m matrix; equal trailing nodes are matched before it is built, which leaves the matrix small when a file changed little (NFR-P6).

Concurrency

ID	Requirement	Source
NFR-C1	Repository data shall be safe to read and write from background threads while the GUI reads it.	History f03347d4, 3d84bc03, 7fde1962, 04825b26
NFR-C2	Listener events shall reach every listener even if one throws or events fire from a background thread.	History (CopyOnWriteArrayList, per-listener isolation)
NFR-C3	The GUI shall keep repository writes off the JavaFX thread and UI reads on it.	History 5cb865c3, 56a51e6a, dae048f6

Compatibility and evolution

ID	Requirement	Source
NFR-V1	A repository in an older or retired format shall be refused with an explanation, never fail with an exception or be misread.	Code `ArtifactData#retiredFormat`, tree-format checks; History ff3c5dfe
NFR-V2	Where an adapter's tree format changed, old repositories should stay checkout-only rather than become unreadable.	History d9ce686c (C++)

ID	Requirement	Source
NFR-V3	Newer builds shall read older supported formats; older builds need not read newer ones. A format change that can be converted shall be converted on the next write, once: a core with a per-artifact index is rewritten without it (FR-P4) - measured on a 15-commit repository (54 MB core, 15 packs), its 16th commit took 121 s and left a 49 KB core and 1 pack, with identical checkouts before and after. A build from before that change refuses the new core as “an older storage format”.	Test `PackOrderMigrationTest`; History 81442a9e, 7c18da73; uncommitted 2026-10-13

Security

ID	Requirement	Source
NFR-S1	The sync server shall accept only local connections unless explicitly started on all interfaces.	Doc
NFR-S2	The REST server shall reject upload and repository names that escape their storage folder.	History ad74aaad
NFR-S3	Remote sync is unauthenticated; the GUI shall ask for confirmation before accepting connections from other machines.	Doc; Code `ServerView` - the missing authentication is a stated limitation
NFR-S4	The REST server shall sign its tokens and refuse unsigned or wrongly signed ones.	Test `RestTokenSecurityTest`, `RestShippedConfigTokenTest`
NFR-S5	Outside development mode the REST server shall refuse to start without a JWT secret of at least 32 characters other than the default, and a users file holding password hashes.	Test `RestSecurityTest`

Platform and build

ID	Requirement	Source
NFR-B1	The system shall build and test with JDK 21+ and the bundled Gradle wrapper, with no other prerequisite.	Doc
NFR-B2	Tests that need optional Python modules (`libcst`, `tree-sitter`, `py4j`, `lymodel`) shall be skipped, not failed, when the modules are absent - decided by the adapters’ own interpreter lookup (AD-9), so a test runs whenever the adapter could.	Doc; Test `Lymusic`, `PythonAvailable`
NFR-B3	CLI and REST shall run headless, without JavaFX on the classpath.	History bfcddc23
NFR-B4	The GUI shall be installable as a native application on macOS, Linux and Windows.	Doc (jpackage tasks)
NFR-B5	Third-party dependency versions shall be kept in one version catalog.	Doc (`gradle/libs.versions.toml`)

Known gaps

Requirements the code does not meet, places where the documentation and the code disagree, and partial implementations. Ordered by how likely they are to surprise a user.

#	Gap	Where	Kind
1	The README maps `*.c/*.h/*.cpp/*.hpp` to the C and C++ adapters by default, but both are disabled by default, so those files go to the File adapter (whole-file granularity, no VEVOS traces) unless enabled in Preferences.	`AdapterPreferences.DEFAULT_DISABLED_PLUGIN_IDS` vs README	Doc/code mismatch - **fixed 2026-10-03**: C and C++ on by default (new repositories only)
2	The CLI only works from the repository root: it fixes the repository at `./.ecco`, though the README promises a search of parent directories (`EccoService.detectRepository` exists but is not used).	`cli/.../Main.java:41` (IF-C8)	Doc/code mismatch - **fixed 2026-10-03**: commands search upwards; the repository's directory, not the current one, is the working directory (README corrected)
3	`.hashes` is written on checkout but never used: the unchanged-file check is commented out, so every commit re-reads every file.	DR:469-476 (FR-M11)	Partial, **done another way 2026-10-13** - measured 2026-10-03, README corrected: reading is 4.5% of a commit for line-based files but 69% for Java AST files (about 1.7 ms per file). 2026-10-13: instead of `.hashes`, which a Git import never has, a file read with the same content and context as in the previous read is not parsed again: its tree is copied (`ArtifactReader#readContext`, `Trees#copyRead`, opt-in for C, C++, TypeScript, Python, Java AST, Markdown, LilyPond). 15 flipperzero-firmware commits: 238 s to 205 s, a typical commit 13 s to 10 s; associations and checkouts identical with and without (`DispatchReaderReuseTest`). `.hashes` itself is still unused
4	The `java` (lines) and `challenge` adapters have no writer: a checkout of their files writes nothing. Both are off by default.	`JavaWriter.java:30` in both adapters	Partial - **fixed 2026-10-02**: both writers now refuse with a message naming the Java (AST) adapter; documented as read-only
5	Several adapters claim `.java` (java-ast, java, challenge, runtime) and `.c/.h` (C, C++) at the same priority; if more than one is enabled, which one wins depends on set order.	adapter patterns	Ambiguous requirement - **fixed 2026-10-03**: ties broken by plugin id and logged; C++ yields `.c/.h` to C
6	Condition minimization is preview-only; it is not applied to checkout.	`minimize-preview`; FR-N7	**Done 2026-10-03 (opt-in)**: checkout can use stored revision-exact minimized conditions (`minimize`, `checkout --minimized`, GUI preference); `MinimizedCheckoutEquivalenceTest` shows identical files for every model-consistent configuration. Measured on 30 generated variants: no gain there (conditions already minimal after absorption), a one-off main tree costs ~13 ms per checkout
7	Partial-order-graph alignment is factorial in concurrent unresolved branches; no bound is enforced beyond a capped fallback.	`PartialOrderGraph`	Performance risk - **characterized 2026-10-03**: the fallback (since July) keeps commits at 10-53 ms with 30 concurrent unique lines; the exact algorithm alone runs out of memory there (`ConcurrentBranchScalingTest` fails without the fallback). Its known under-matching gave 0 wrong files in 990 adversarial checkouts. Exact replacement not needed now; a design is noted in the gap plan. A separate cost, aligning two long paths in an n x m matrix (quadratic in a file's length, for every file of every commit), was cut 2026-10-13 by matching equal trailing nodes first (NFR-P6)

#	Gap	Where	Kind
8	Remote sync is unauthenticated; the REST server has hard-coded demo users with plaintext passwords and a default JWT secret, and its roles are never checked.	README; `rest/.../DummyUserDB`, `application.yml`	Security limitation - **hardened 2026-10-03**: the JWT secret key in `application.yml` was misspelled, so tokens were unsigned and a forged unsigned token was accepted for any user - fixed; the server refuses to start without a real secret and a users file (PBKDF2 hashes) outside development mode; the GUI confirms before syncing with other machines. Still open by choice: sync authentication, role checks
9	The REST storage directory is not configurable; it is chosen by heuristics (Docker, a `jenkins` user, else `examples/`).	`rest/.../Settings`	Missing configuration - **fixed 2026-10-03**: `ECCO_STORAGE_DIR` / `ecco.storage-dir`, heuristics as fallback, logged at startup
10	Java 21 pattern-matching `switch` and record patterns are not supported by the Java (AST) adapter and fail loudly; comments inside lambdas and expressions are unverified.	`JavaASTReader`	Partial - **fixed 2026-10-03**: JavaParser 3.27, Java 21 level (round-trip tests); checking showed comments inside lambdas and expressions WERE lost - now kept as metadata, filled in for old repositories on the next commit
11	Non-lazy composition throws “not yet implemented”.	REPO:831	Not implemented (unused) - **fixed 2026-10-02**: the unused option and its branch removed
12	Unknown `[<id>]` features in a configuration silently create a temporary feature (the strict check is commented out).	`ConfigurationParser:160`	Weak validation - **fixed 2026-10-02**: unknown ids are rejected; fixing it also showed that `[<id>]` and `[<id>].<rev>` of an *existing* feature resolved to a new revision instead of the latest or the given one, fixed too
13	A local fork opens its origin read-write (TODO “init read only!”).	ES:756	Partial - **resolved 2026-10-03**: a fork only takes shared locks and reads its origin in a read-only transaction; `ForkLeavesOriginUnchangedTest` shows the origin is left byte-for-byte unchanged, so no read-only mode is needed
14	GUI leftovers: `ForkView` unused (no Fork in the GUI), `FeatureDetailView` and `PresenceConditionDetailView` unreferenced, several `ArtifactsView` toggles marked TODO.	`gui/...`	Dead / unfinished UI - **fixed 2026-10-03**: Fork is on the Repositories ribbon (rewritten on `EccoService.fork`, test `ForkViewTest`); the dead views and placeholder toggles removed
15	`lilypond-config.properties` contains a hard-coded per-user path.	`adapter/lilypond/src/main/resources`	Portability - **fixed 2026-10-02**: bundled defaults empty, `lilypond` looked up on the PATH
16	`AdapterPreferences.java` holds a raw NUL character in a string literal, so git treats the file as binary and hides its diffs.	`service/.../AdapterPreferences.java:22`	Hygiene - **fixed 2026-10-02**
17	The changelog stops at 0.1.9 and no requirements or release notes record the work since; this document is the only consolidated statement of intent.	`CHANGELOG.md`	Documentation - **fixed 2026-10-03**: a 0.2.0 section with upgrade notes, new features, performance, fixes and security, written from the history
18	The CLI reported nothing on success: `commit` printed no commit id, `checkout` no hint at its warnings, and a commit violating accepted constraints said nothing (only the GUI checked).	`CommitCommand`, `CheckoutCommand`	Missing feedback - **fixed 2026-10-03** (found writing the use cases): `commit` prints the id, the configuration and `CONSTRAINT:` lines; `checkout` prints a count per warning kind (IF-C9)
19	A REST commit with a malformed configuration or none at all failed with 500, after the files were written.	`FileRepositoryService.addCommit`	Wrong status - **fixed 2026-10-03** (use cases): refused with 400 before anything is written (IF-R3)
20	The sync server’s log in the GUI showed empty rows: the message column always rendered “” and the time column had no value. Reopening the view while a server runs still shows “port -1”.	`ServerView`	UI bug - **fixed 2026-10-03** (use cases), test `ServerViewLogTest`; the port -1 header **fixed 2026-10-03** too (the service reports the bound port)

#	Gap	Where	Kind
21	One failed commit ends a Git import run (e.g. a blank configuration on a tree without `.config`); the commit cannot be corrected and retried.	`ImportGitView.reportImportFailure`	Usability - **fixed 2026-10-03**: the failed commit's review screen comes back with the error and the configuration tried, for Import again or Skip, also for an unattended auto-import; test `ImportGitViewRetryTest` (FR-G4, UC-10 SA7)
22	Accepted constraints are stored in the repository and travel with fork and pull; rejected ones are only stored in the local preferences, so collaborators are offered suggestions someone else rejected.	`ConstraintSuggestionsView`, `ConstraintSuggestionPreferences`	Inconsistent design - **fixed 2026-10-03**: rejections are stored in the repository next to the accepted constraints and travel with fork/pull/push; a merge keeps the receiving repository's own decisions; local rejections move into the repository when the Feature Model view first loads; older builds ignore the new field; test `RejectedConstraintsTest` (FR-N4)
23	Resolving an ORDER warning commits the whole checkout directory under the checkout's configuration, so the checkout's SURPLUS content and MISSING gaps are counted as that configuration too.	`CheckoutDetailView.applyFix`	Data-quality risk - **fixed 2026-10-03**: first pinned down (`OrderResolutionCharacterizationTest`: the commit hid the configuration's MISSING and SURPLUS warnings; committing only the reordered file instead would drop every other file from that configuration). Reorder… now records the order in the artifact's partial order graph without a commit (`EccoService#recordOrder`, FR-K11, UC-6); on the command line `order <file>...` (IF-C10)
24	FR-D2 said fork goes "into an empty location"; the code only refuses a location that already holds a repository.	ES:666-668	Doc/code mismatch - **fixed 2026-10-03**: FR-D2 corrected
25	The front-end matrix showed the CLI's constraint support as "suggest + preview"; it can also `minimize`, but cannot accept constraints.	matrix	Doc - **fixed 2026-10-03**
26	`fetch` also stores the remote's features with the remote's entry, which FR-D3 did not say.	`RemoteSyncService.fetch`	Doc - **fixed 2026-10-03**: FR-D3 extended
27	An unknown remote was reported as "Remote 'x' does not exist." by fetch and pull but "Remote x does not exist" by push.	`RemoteSyncService.push`	Inconsistency - **fixed 2026-10-03**, test `RemoteSyncCharacterizationTest`
28	Local fetch and pull still open the other repository with `// TODO: init read only!`; #13 settled this for fork only.	`RemoteSyncService`	**Resolved 2026-10-03**: `FetchPullLeaveRemoteUnchangedTest` shows a local fetch, pull, pull with exclusion and failed pull leave the remote byte-for-byte unchanged, so no read-only mode is needed; the TODOs replaced by that note (FR-D7)
29	A whitespace-only line right after a Markdown list was lost on commit, together with the empty lines before it: CommonMark counts it as part of the list's span, after its last item, and no block claimed it.	`MarkdownTreeBuilder.translateChildren`	Data loss - **fixed 2026-10-04**, found by the generated round trips of `MarkdownRoundTripPropertyTest` (2,000 documents); files committed before keep the old tree, the lines come back from the next commit on
30	The GUI could delete any existing directory: New offered "A repository already exists … Delete it and start fresh?" for any existing path - a working directory typed instead of its `.ecco`, or the GUI's current directory for an empty field - and deleted it on Yes. Open failed silently on an empty or relative path.	`InitView`, `OpenView`	Data loss - **fixed 2026-10-04** (headless GUI tests): only a directory named `.ecco` is replaced, relative paths are refused
31	"Delete contents?" before a checkout kept every file whose absolute path contained ".ecco", so a checkout directory under e.g. `~/.ecco-work` kept its old files.	`DeleteDirectoryVisitor`	Bug - **fixed 2026-10-04**: only a `.ecco` below the directory is kept
32	Checkout: answering No to a question went back to an empty form; pressing Checkout again checked out an empty configuration into the repository's home directory.	`CheckoutView`	Bug - **fixed 2026-10-04**: the form keeps what was entered
33	Variants: checking out several variants put them all into one folder named "Commit" (every committed variant's name); the second failed and left the toolbar disabled.	`VariantsView`	Bug - **fixed 2026-10-04**: one folder each, by id where names collide

#	Gap	Where	Kind
34	Commit comparison showed associations of only one commit under the other commit's columns.	`CommitComparisonView`	Bug - **fixed 2026-10-04**
35	Minimized conditions invalidated by a commit stayed visible in the GUI (it waited for an event the service no longer fires).	`MinimizationResults`	Bug - **fixed 2026-10-04**
36	The pull and push dialogs listed every remote again each time the user went back to the remote step.	`PullView`, `PushView`	Bug - **fixed 2026-10-04**
37	A failed local pull or push reported "Error rolling back transaction: No transaction active" instead of its cause (e.g. an unknown feature in the exclusion).	`RemoteSyncService`	Bug - **fixed 2026-10-04**: rollbacks only when a transaction is active
38	Smaller GUI issues found by the view tests: Cancel while committing did nothing (there was no commit cancellation); Commit Multiple Versions left the working directory at the last committed folder; the pull dialog's feature tree was empty until the remote was fetched once; Remove Variant could leave the toolbar disabled if removing threw; the commit dialogs checked constraints after committing, in a modal dialog from inside the commit - a check that can never find anything, since only exception-free rules count and the committed configuration is the exception to every rule it violates.	`CommitView`, `CommitBaseDirView`, `PullView`, `VariantsView`, `CommitService`, `ImportGitView`, `CommitCommand`	Usability - **fixed 2026-10-11**: `EccoService#cancelCommit()` stops a commit before it stores anything (per file read, before extracting, before storing) and rolls it back; Commit Multiple Versions commits no further folder and restores the working directory; the pull dialog fetches in the background; Remove Variant re-enables in `finally`; Commit, the Git import and the CLI's `commit` check constraints before committing, like Commit Multiple Versions
39	A fork over the network takes the new-repository settings (`.ecco/.settings`), not its origin's: the pull protocol does not carry them. Forking a repository from before musical tokens (or variable contexts) this way reads its LilyPond files with them, which never match its artifacts; one read with parce gets `lilypond.parser=lybar-1` this way and takes commits it should refuse (AD-10). A local fork copies the origin's settings.	`EccoService.fork(String, int, String)`	Gap - noted 2026-10-08; **fixed 2026-10-14**: the server answers a `SETTINGS` request and a fork over the network takes its origin's settings. Writing the test showed that a fork over the network had never worked (the copy needed the entity factory before `init()` made it). Pull and push check settings too (FR-D8)
40	The LilyPond reader fails on one vocal score of the silverswan corpus: `NullPointerException` in `LilyEccoTransformer.transformVariableDefinitonNode` (a context node where a token was expected). With and without musical tokens; found reading the 665 corpus files.	`LilyEccoTransformer`	Bug - noted 2026-10-08; not reproduced 2026-10-11: all 724 corpus and example files commit, with lymodel's parse and with parce. **Reproduced 2026-10-13 in plain-token mode only** (without musical tokens), on 4 corpus files - see #51; **fixed 2026-10-14** with #51
41	Commit Multiple Versions showed a `.config` written one feature per line (with blank lines) as ragged multi-line rows, in both the configuration and the default commit message.	`CommitService.getConfigStringFromFile`	Usability - **fixed 2026-10-08**: line breaks collapsed into one line (FR-M3)
42	The Python and LilyPond adapters ran plain `python` from the PATH: a GUI commit of a musical LilyPond repository failed with "No module named 'lymodel'" although another python had it, and a GUI started from the Finder (no MacPorts or Homebrew on its PATH) would find no python at all. The tests checked `python3` instead, so they skipped where the adapter could read.	`FileParser`, `PY4JParser`, `Lymusic`	Bug - **fixed 2026-10-08**: interpreter lookup (AD-9), tests ask the same lookup (NFR-B2)
43	Six LilyPond regression tests failed without lymodel instead of being skipped, and a lymusic test expected v6's ties in checkouts without `slurs.1` - the ties arrive with slurs in the example history, so ECCO's checkout was right. Both went unnoticed because the music tests only run with lymodel.	`service` tests	Test defects - **fixed 2026-10-08** (5d0e7ff1, 68ceeecf)

#	Gap	Where	Kind
44	Written back after a commit, 71 of 724 LilyPond corpus and example files hold other music than they were read with (lymusic: notes, lyrics, articulations, slurs, dynamics or attributes) - the same 71 with lymodel's parse and with parce, mostly with the same first difference, so in the shared transformer or writer.	`LilyEccoTransformer`, `LilypondWriter`, `LilypondFormatter`	Bug - noted and fixed 2026-10-11: the writer ran numbers and signs together that were apart in the file read (`\shp 0 4` as `\shp 04`, `a2 2` as `a22`, `^ ^` as `^^`) - tokens now keep together exactly what was written together, and never run into one word otherwise; and a `\relative` without a pitch of its own was spelled from the note before it, not from f. All 719 lymusic reads come back with their music - measured after a single commit: the positions that decided what was written together came from the commit that stored each token, so a checkout combining commits ran tokens together or split them (`\p e` as `\pe`, `e2 f4` as `e2f 4`, a code review found it the same day). Fixed with `lilypond.parser=lybar-2`: a token records whether it was attached to the one before it where a space would change the music, as part of its identity (`LilypondSpacingAcrossCommitsTest`).
45	One commit of all 724 LilyPond corpus files ran out of memory with a 6 GB heap, writing the artifact pack; in batches of 40, one repository each, they commit.	`SerTransactionStrategy.writePack`	Gap - noted 2026-10-11. Since 2026-10-13 such a commit rolls back and the command line reports running out of memory with the heap size and how to raise it (FR-M10, IF-C3); before, it showed only "Not all transactions have been ended". The pack is still built in memory
46	A review of the parce removal found the tree-sitter parse reading markup and modes by its flat tree: in `\markup \bold { ... }` and `\drums { ... }` the braces are no child of the command, so markup words were read as notes, slurs and variables (a variable resets the pitch `\relative` spells the next note from) and drum notes as variables; a chord's or tremolo's `:` as a script, the chord steps after it as variables; `#:kw` as two words written apart (`#: kw`, broken Scheme); an empty file or a brace tree-sitter supposes missing as an empty token; text tree-sitter left out as one token with its line breaks; positions in code points where Java counts UTF-16; the parser's warnings dropped when it succeeded (a file read without musical tokens said nothing). And a repository without LilyPond settings - every one from before them, and every one ecco-rest made without the LilyPond plugin - refused its first LilyPond commit though it held no LilyPond to disagree with.	`LilypondParser_2.py`, `FileParser`, `EccoService`	Bug - **fixed 2026-10-11** in `lybar-2`: markup and drum or chord mode are spans after their command (markup's end from lymodel's `markup_span`), markup read by words; `:` a chord separator; Scheme keywords and characters one token; no empty tokens; left-out text by words; UTF-16 positions; warnings logged; a plugin's settings are adopted by a repository with none of them and none of its files (AD-10). Open: pull does not compare settings (a remote read with other settings merges tokens that never match).
47	Only the commit rolls its transaction back on an `Error` (such as `OutOfMemoryError`); 16 other service operations that open a read-write transaction roll back only on an `Exception`, so an `Error` leaves their transaction open and closing the repository fails afterwards.	`EccoService` and the services under `service/.../service/`	Gap - noted 2026-10-13 (found fixing the commit, 2dc75b70); **fixed 2026-10-14**: all 28 places that open a transaction, read-only ones too, roll back on an `Error` and pass it on; a failed init or fork removes the new repository also then. Worse than noted: transactions nest by a counter, so every later operation became part of the open one and its writes were never stored (`ErrorRollbackTest`)
48	Every association a commit touches is written again, because its counters change with every commit; on flipperzero-firmware that is about 14% of the time spent storing. Its artifact tree, which changes far less often, could be stored apart from the counters.	`SerTransactionStrategy.endReadWrite`	Performance - noted 2026-10-13
49	To tell whether an artifact changed, a commit serializes every artifact it touched and compares the digest with the stored one - about half of the time spent storing after the 2026-10-13 storage changes, though most of those artifacts are unchanged.	`SerTransactionStrategy.writeArtifactPack`	Performance - noted 2026-10-13

#	Gap	Where	Kind
50	7 of ECCO's own Java files cannot be read by the Java (AST) adapter. Since AD-11 they are committed by the text adapter (lines, byte-exact) with a warning, not refused.	`JavaASTReader`	Bug - noted 2026-10-13; **fixed 2026-10-14**: of all 1,262 Java files in the repository, 8 failed - 7 test inputs that are not valid Java on purpose (fragments, `abstract private`), which belong to the text adapter, and `TextFileFormat`, which imported two classes twice: the imports are an unordered set of children, which refused two equal ones. A repeated import is now read once (`JavaASTDuplicateImportTest`). Two identical initializer blocks in one class still fail the same way and go to the text adapter
51	In plain-token mode (a LilyPond repository without musical tokens) the reader throws a `NullPointerException` on 4 corpus files - #40's failure, which does not occur with musical tokens. Since AD-11 such a file is committed by the text adapter with a warning, so a commit no longer fails, but its music is traced by lines.	`LilyEccoTransformer`	Bug - noted 2026-10-13; **fixed 2026-10-14**: the 4 files name the voice of `\lyricsto` without quotes (`\lyricsto Alto`), a token of its own, where `transformLyriclist` always took two steps down to a string. Now all 705 corpus files are read and written back byte-identical in both modes (`LilypondParseTest`)